

# 视觉组 颜铭含 12月工作月报

## 一、工作流程

OpenCV 学习 → YOLO 目标检测 → Ubuntu 环境配置 → Gazebo/Isaac Sim 仿真 → ROS2 学习

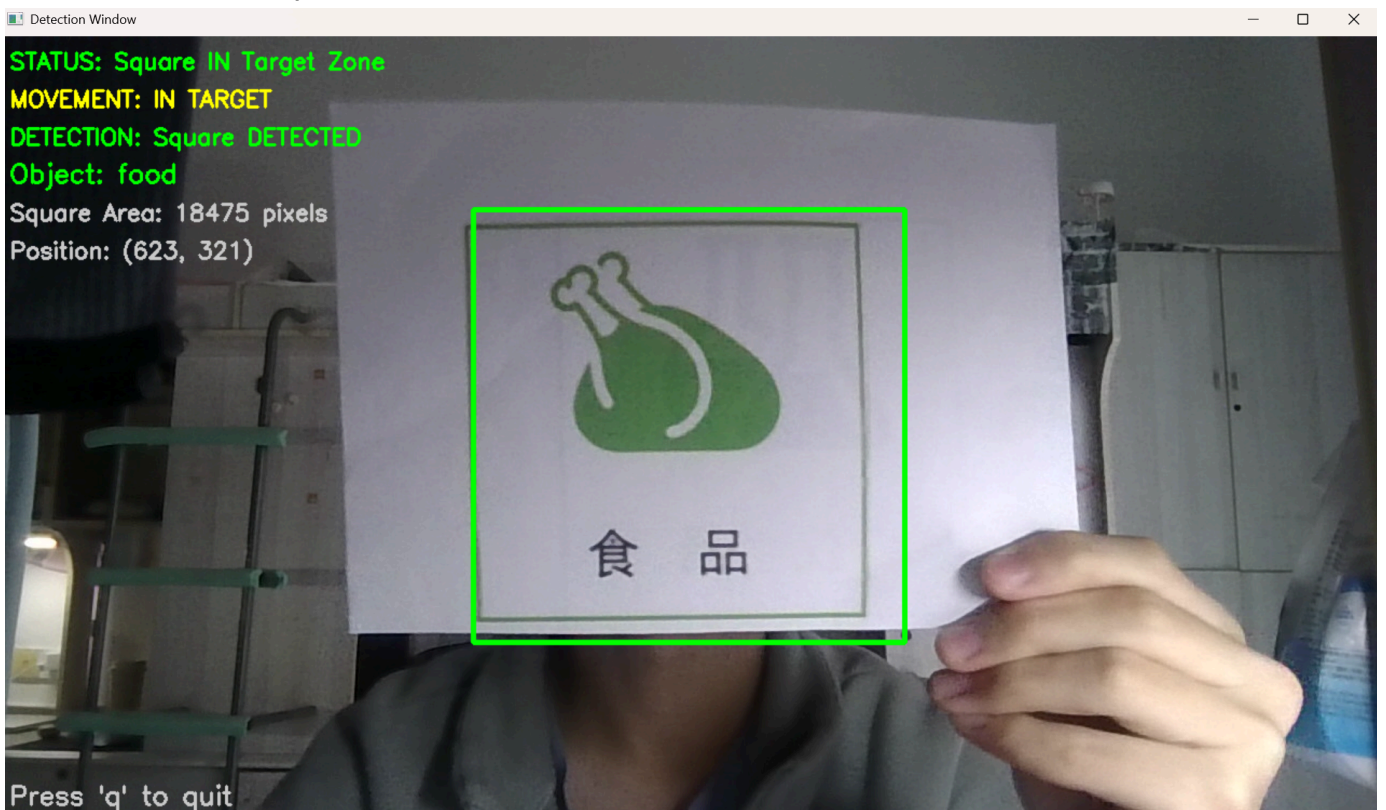
## 二、工作内容

### 0. 技术路线确定

- Unitree L2 (原始点云) → FAST-LIO2 (实时里程计/建图) → Pointcloud\_to\_Laserscan (降维处理) → Nav2 (全局/局部路径规划)

### 1. OpenCV

- 学习重点：** 图像预处理（灰度化、高斯滤波）、色彩空间转换（HSV/RGB）、边缘检测等
- 具体应用：** 实现基于传统视觉的简单目标提取
- 产出：** 编写了使用OpenCV的识别模块



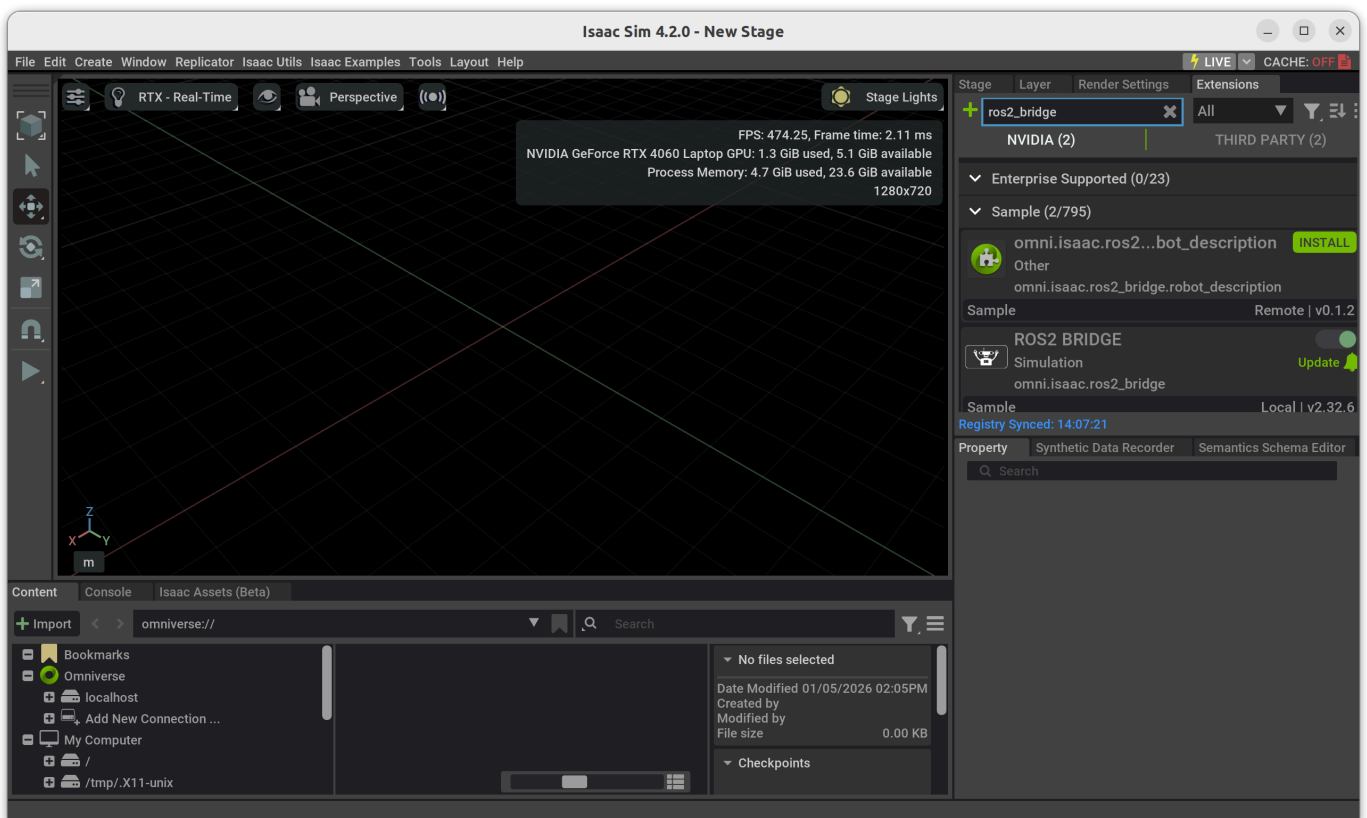
### 2. YOLO

- 任务内容：** 学习使用 YOLO v8，用于赛场复杂场景下的目标识别，此工作后移交黄益鹏负责
- 实施细节：**

- **数据集准备：** 进行数据采集与标注（XAnyLabeling / Roboflow）
- **模型训练：** 针对比赛目标进行迁移学习，优化训练参数，开始尝试TensorRT加速

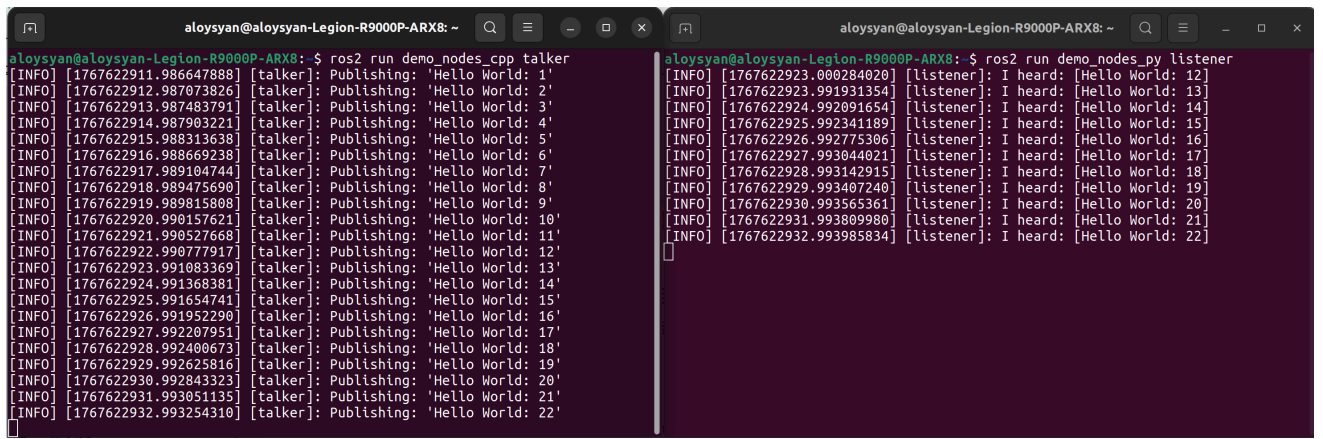
### 3. 仿真

- **系统部署：** 配置 Ubuntu 开发环境，配置 NVIDIA 相关驱动与 CUDA 环境，加装CuDNN卷积神经加速模块
- **仿真搭建选型：**
  - **Gazebo：** Gazebo Harmonic
  - **NVIDIA Isaac Sim：** 4.2.0
- 现在 Gazebo 上部署，后视情况观察是否需要移植到 Issac Sim



### 4. ROS2

- **技术选型：** ROS2 Humble
- **开发内容：**
  - 完成环境配置与安装，调试发布了 Talker 和 Listener 话题，待正式开展工作



The image shows two terminal windows side-by-side, both running on a system named 'aloysyan@aloysyan-Legion-R9000P-ARX8'. The left window shows the output of the command 'ros2 run demo\_nodes\_cpp talker', displaying 22 messages from a 'talker' node publishing 'Hello World: 1' through 'Hello World: 22'. The right window shows the output of the command 'ros2 run demo\_nodes\_py listener', displaying 22 messages from a 'listener' node receiving 'Hello World: 12' through 'Hello World: 22'. Each message line includes a timestamp, a log level (INFO), a node name, and the message content.

```
aloysyan@aloysyan-Legion-R9000P-ARX8: ~$ ros2 run demo_nodes_cpp talker
[INFO] [1767622911.986647888] [talker]: Publishing: 'Hello World: 1'
[INFO] [1767622912.987073826] [talker]: Publishing: 'Hello World: 2'
[INFO] [1767622913.987493791] [talker]: Publishing: 'Hello World: 3'
[INFO] [1767622914.987903221] [talker]: Publishing: 'Hello World: 4'
[INFO] [1767622915.988313638] [talker]: Publishing: 'Hello World: 5'
[INFO] [1767622916.988669238] [talker]: Publishing: 'Hello World: 6'
[INFO] [1767622917.989104744] [talker]: Publishing: 'Hello World: 7'
[INFO] [1767622918.989475690] [talker]: Publishing: 'Hello World: 8'
[INFO] [1767622919.989815808] [talker]: Publishing: 'Hello World: 9'
[INFO] [1767622920.990157621] [talker]: Publishing: 'Hello World: 10'
[INFO] [1767622921.990527668] [talker]: Publishing: 'Hello World: 11'
[INFO] [1767622922.990777917] [talker]: Publishing: 'Hello World: 12'
[INFO] [1767622923.991083369] [talker]: Publishing: 'Hello World: 13'
[INFO] [1767622924.991368381] [talker]: Publishing: 'Hello World: 14'
[INFO] [1767622925.991654741] [talker]: Publishing: 'Hello World: 15'
[INFO] [1767622926.991952290] [talker]: Publishing: 'Hello World: 16'
[INFO] [1767622927.992207951] [talker]: Publishing: 'Hello World: 17'
[INFO] [1767622928.992400673] [talker]: Publishing: 'Hello World: 18'
[INFO] [1767622929.992625816] [talker]: Publishing: 'Hello World: 19'
[INFO] [1767622930.992843323] [talker]: Publishing: 'Hello World: 20'
[INFO] [1767622931.993051135] [talker]: Publishing: 'Hello World: 21'
[INFO] [1767622932.993254310] [talker]: Publishing: 'Hello World: 22'

aloysyan@aloysyan-Legion-R9000P-ARX8: ~$ ros2 run demo_nodes_py listener
[INFO] [1767622923.000284020] [listener]: I heard: [Hello World: 12]
[INFO] [1767622923.991931354] [listener]: I heard: [Hello World: 13]
[INFO] [1767622924.992091654] [listener]: I heard: [Hello World: 14]
[INFO] [1767622925.992341189] [listener]: I heard: [Hello World: 15]
[INFO] [1767622926.992775306] [listener]: I heard: [Hello World: 16]
[INFO] [1767622927.993044021] [listener]: I heard: [Hello World: 17]
[INFO] [1767622928.993142915] [listener]: I heard: [Hello World: 18]
[INFO] [1767622929.993407240] [listener]: I heard: [Hello World: 19]
[INFO] [1767622930.993565361] [listener]: I heard: [Hello World: 20]
[INFO] [1767622931.993809980] [listener]: I heard: [Hello World: 21]
[INFO] [1767622932.993985834] [listener]: I heard: [Hello World: 22]
```

### 三、1月计划

1. **算法优化：** 在 Nanodet 上进行 TensorRT 加速
2. **仿真：** 视觉仿真部署
3. **ROS2：** 与电控组配合，学习通信API接口编写